

WORKSHOP: "ADVANCING HEALTHCARE DEVELOPMENT IN INDIA:
INTEGRATING PHARMACEUTICALS AND IPRS FOR GLOBAL
COMPETITIVENESS"

BHOPAL, INDIA, JAN. 10, 2025

“DRUGS: THE GOOD, THE BAD, & THE UGLY”

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Agenda

Introduction: Pharmaceuticals and Patents

Developing Country-specific Issues

Engaging the audience:

- “Evergreening”
- Covid-19 and TRIPS Waiver
- Vaccine Nationalism

PHARMACEUTICALS & PATENTS



Many Ways to Appropriately the Returns to Innovation:

- Trade Secrecy
- Lead Time Advantages
- Reputation
- Capital Barriers

Pharm is most sensitive (due to easy of copying drug innovations)

Factors:

- R&D expensive, risky
- Clinical trials
- Global free-riding
- Subject to regulation – drugs are important to Public Health!

Pharmaceuticals dependent on Patent Rights

Other realities

Pharm firms depend on Patents, Brand name, and Data Exclusivity

- ❑ But never full 20 years due to marketing authorization & approvals
- ❑ Market power, high prices, access are public concerns. In the background, firms face:
 - Threats of Compulsory Licensing
 - Price Controls
 - Limited Injunctive Relief
 - Parallel Trade

How to Mitigate Market Power & Markup Pricing

International Price Discrimination

(works if you prevent resale)

Mutual interest (North & South). Why?

How to Mitigate Market Power & Markup Pricing

Compulsory licensing (CL) if Voluntary licensing (VL) not successful

Government use (health emergencies)

Price controls

Generic competition (once patents expire)

Developing Countries' problem: Access

Access depends on many factors: poverty, taxes/tariffs, infrastructure, supply chains, corruption, bureaucracy, regulations

Private sector Pharmaceutical R&D targeted to rich consumer diseases.

- Focused on short run projects (immediate needs, short life expectancy) – not long-term projects (taking years to develop)

Other diseases not amenable to a Patent/Market-driven system: **NEGLECTED TROPICAL DISEASES** due to limited purchasing power/market problems.

- Alternatives: public R&D, advance commitments, prizes, subsidies

Developing Countries' problem: Access

1. Pharmaceutical companies “game the system” – via incremental innovations that “extend” their monopoly right over drugs ([Evergreening](#))
2. Supply deficiencies – lack of indigenous innovation and production capacities. TRIPS flexibilities not timely and effective ([Case of Covid-19](#))
3. Need for an improved Global Facility (COVAX-like) and *International Pandemic Treaty* for Future contingencies ([Combat Vaccine Nationalism](#))

CONTROVERSIES

EVERGREENING

Economics of I-MAK (“Initiative for Medicines, Access, and Knowledge”)

Blame high drug prices on patent thickets

Claim pharmaceutical firms “set prices at will”

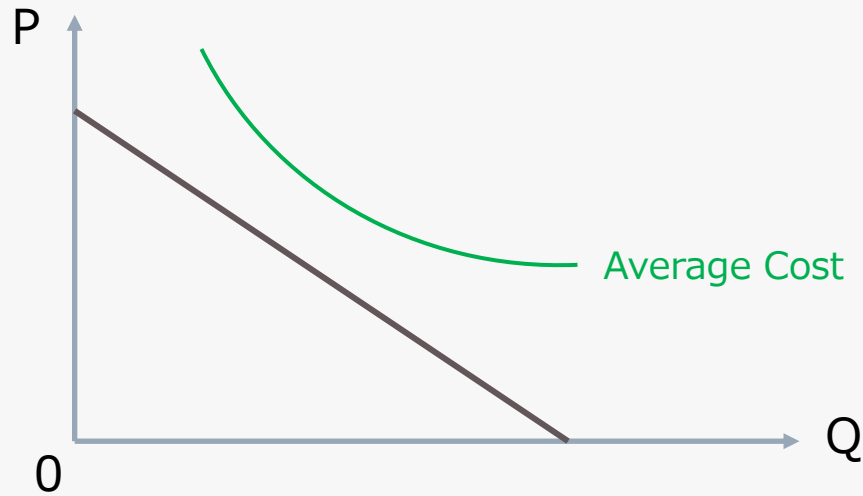
Criticisms:

- Economic arguments lack solidity
- Evidence is lacking

Two versions: contemporaneous and sequential patenting

EVERGREENING

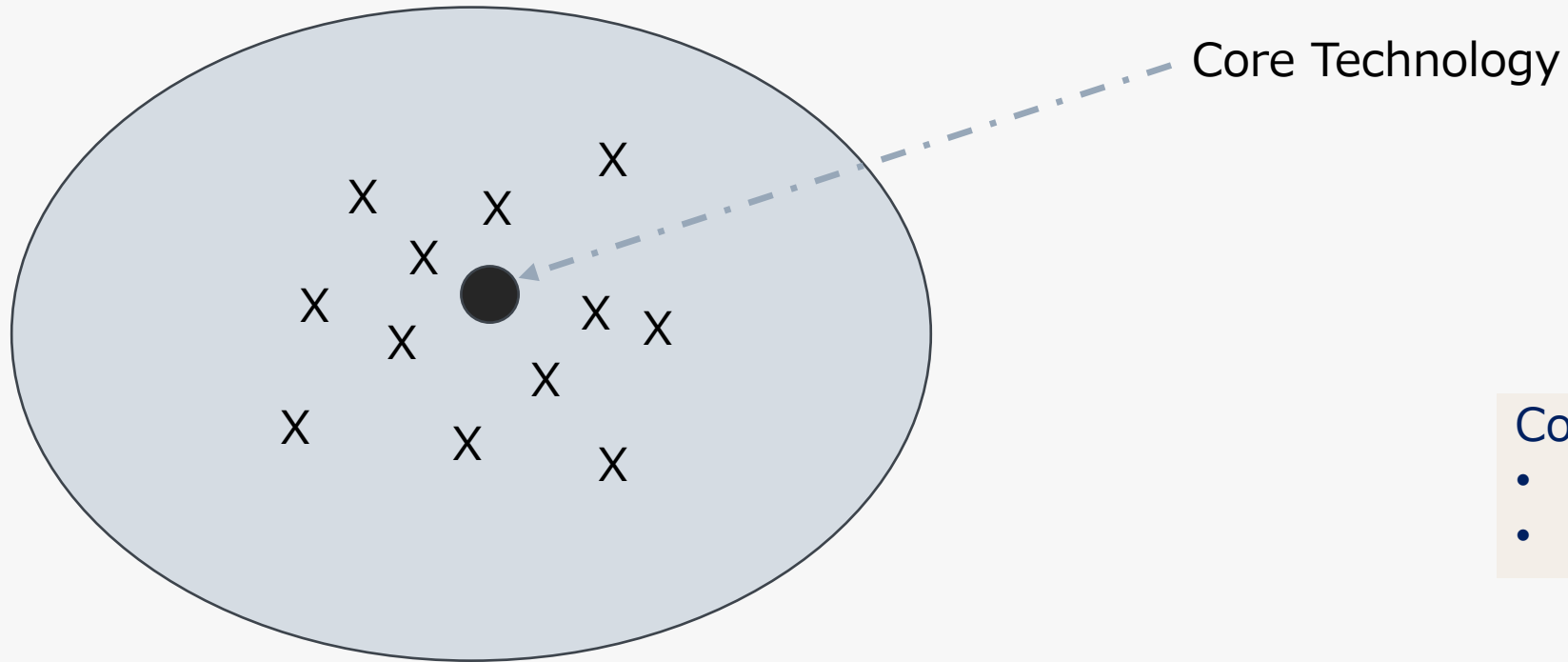
No monopolist can price at will



Monopolists can technically earn negative economic profits

EVERGREENING

Contemporaneous Patenting ("thickets" idea)



Costly strategy:

- Each patent needs fees
- Marketing approval?

Industry Group	Mean
Automobiles & components	52.2
Capital Goods	31.6
Commercial & Prof. Services	5.4
Consumer Durables & Apparel	48.1
Consumer Services	6.2
Diversified Financials	4.7
Energy	21.7
Food, Beverage, and Tobacco	4.2
Health Care Equipment and Services	16.7
Household and Personal Products	21.2
Materials	13.2
Media and Entertainment	24.3
Pharmaceuticals, Biotech, and Life Sci.	10.3
Real Estate	3.9
Retailing	39.1
Semiconductors & Equip.	40.5
Software and Services	59.0
Technology Hardware and Equip.	60.7
Telecommunication Services	45.8
Transportation	3.9
Utilities	1.5
Overall Average	32.7

PATENT GRANTS BY FORTUNE 500 FIRMS

Average quarterly, 1990 – 2021

Source: Patents View, USPTO

Grants	I-MAK claims:	USPTO (actual)
Lyrica	68	3
Xarelto	32	5

EVERGREENING

Sequential Filing (New Uses of existing innovations)



- Note T not necessarily 20 years (due to marketing approval lags)
- Some incremental innovations are good: provide (marginal) varieties, suitable for different populations
- Normal for some products to be covered by multiple inventions (smartphones, baby diapers)

Many kinds of patents protect drugs: active ingredient, formulation, dosage, method of administering, manufacturing process. A single product can contain different *inventions*.

1. Patents do expire. Some enjoy some term restoration.
2. New competitors (other than generics, biosimilars) can offer a different, abbreviated version than the new-uses patent (as long as non-infringing).
3. Substitutability exists (among therapeutic equivalents)

Other Myths

TRIPS WAIVER

Combating COVID and developing Vaccines occurred under a compressed timeline.

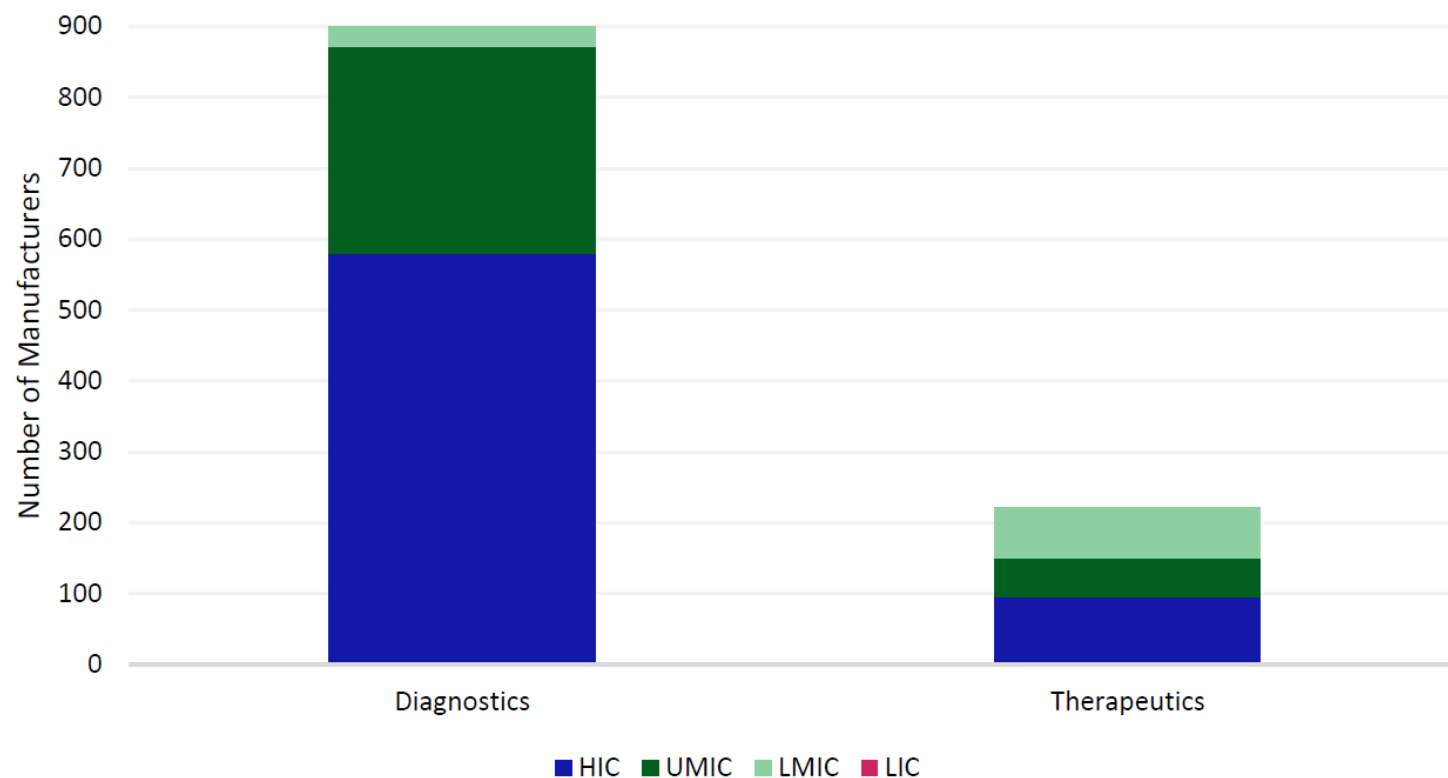
India and S. Africa initiative, late 2020, anxious over supply of vaccines:

- Shortcomings with TRIPS flexibility provisions (e.g. CL) – local manufacturing capacities
- Voluntary licensing (VL) limited to High income country producers.
- Vaccine shortages in developing/least developed countries & **disparities** with developed countries ([Vaccine Nationalism?](#))

TRIPS WAIVER

Figure ES.3 Count of manufacturers of COVID-19 diagnostics and therapeutics

In number of manufacturers. HIC = high-income countries; UMIC = upper-middle-income countries, LMIC = lower-middle-income countries; LIC = low-income countries. Underlying data for this figure can be found in appendix J, [table J.3](#).



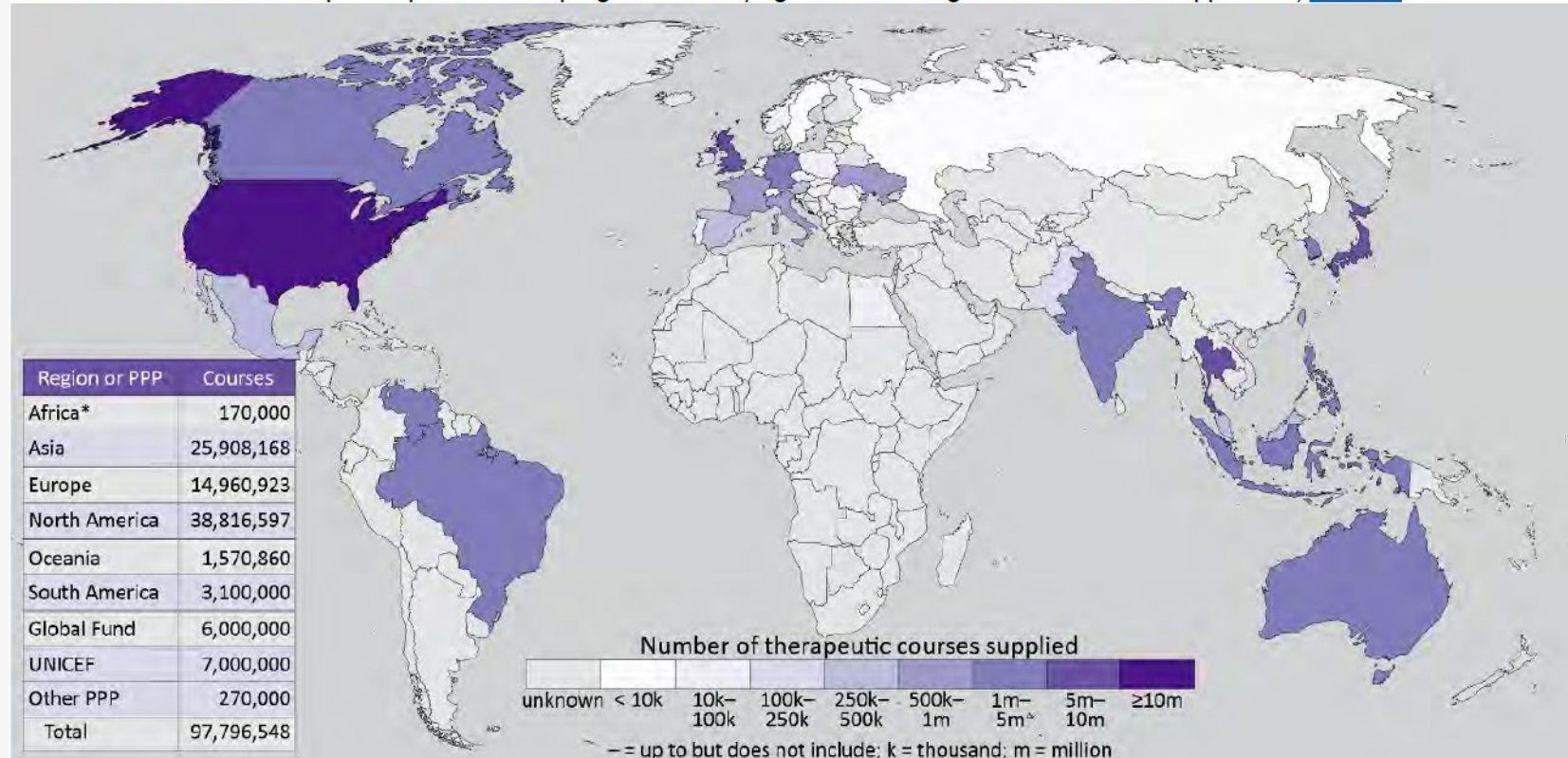
Source:

U.S. ITC (2023)

TRIPS WAIVER

Figure ES.4 Courses of COVID-19 therapeutics made available through donation or purchase, by country and region, based on publicly announced supply agreements

In number of courses. PPP = pooled procurement program. Underlying data for this figure can be found in appendix J, [table J.4](#).



Source:

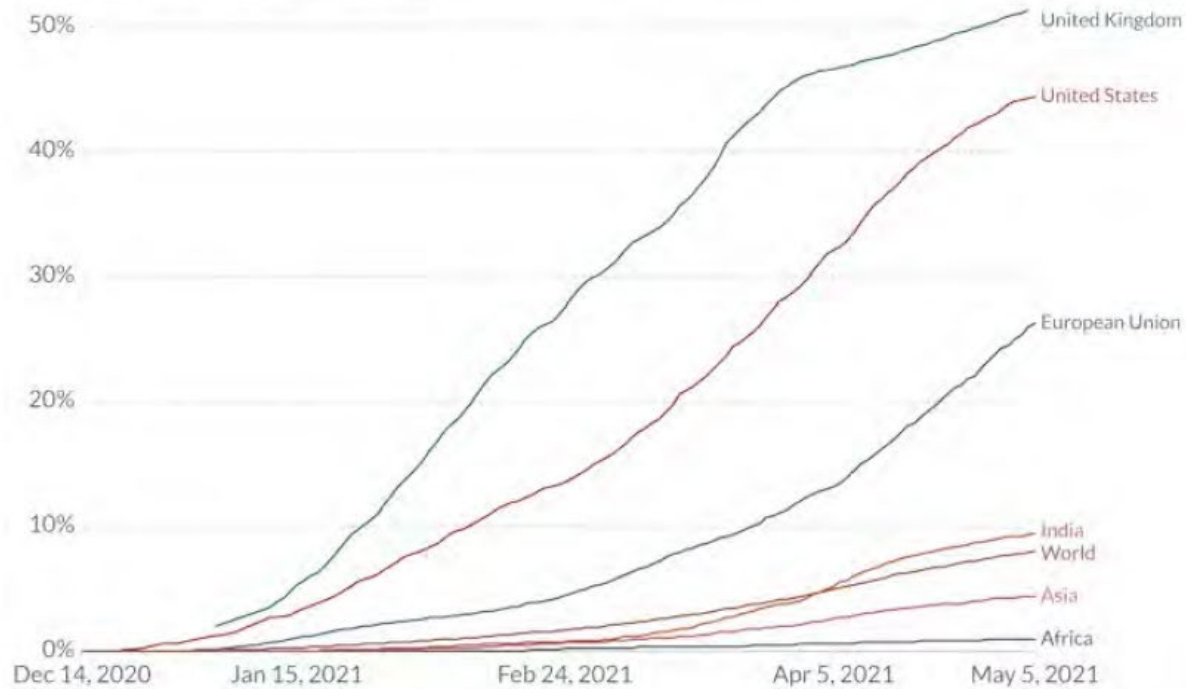
U.S. ITC (2023)

TRIPS WAIVER

Share of people who received at least one dose of COVID-19 vaccine

Share of the total population that received at least one vaccine dose. This may not equal the share that are fully vaccinated if the vaccine requires two doses.

Our World
in Data



Source: Official data collated by Our World in Data

CC BY

Our World in Data, CC BY

Source:
F. Bokhari,
The Conversation,
May 6, 2021

TRIPS WAIVER

Was the “Waiver” Justified?

- Prices not a barrier – the supply chain (raw materials) – world caught off guard.
- Covid tests were off-patent.
- Vaccines weren't the only issue. Masks, ventilators, gloves in limited supply. Nurses and physician shortages.
- Even with waiver, drugs needed to be tested for safety and efficacy.
- Waiver discourages future innovation and preparedness
- Waiver sends bad signal. Undermines TRIP flexibilities. Gives impression they're only for public health emergencies.

TRIPS WAIVER

- Even with waiver and local production (or importing from abroad), prices may not decline as much:

$$TC = VC + FC + FC'$$



Fixed R&D cost = 0 for imitator or copier, but other FC need to be incurred (factory, etc.)

- Lastly, the waiver took a long time (2 years). By late 2022, the pandemic lessened, demand waned, herd immunities developed. Process too slow.

Vaccine Nationalism

During 2021, Supply >> Demand in US, Canada, UK, EU, Australia

Supply $\approx$ Demand in S. Korea, Taiwan, Brazil, Mexico

Supply < Demand in Rest of Latin America, India, Vietnam, China, Egypt

Real or imagined?

Hoarding or geography? Difficult to say because innovators and producers are in developed countries.

Counterproductive – international negative externality – allows variants to emerge. Need a balanced global vaccination. “None of us is safe until we are all safe.” – UN Secretary General, 2020.

Duplicate R&D. Patent races. Careless clinical trials.

Global and local GDP affected.

Lesson learned: International Coordination is required.

Vaccine Nationalism: harmful